



Task 2
Answer Sheet

**Chemistry, Physics, Biology – Tied by
Time and Change**

EOES2025, Zagreb, Croatia
26.04. – 03.05.2025.

Team _____

Students _____



Problem 1. Kinetics of the reaction between crystal violet and hydroxide ions

(TOTAL MARKS: 41.5)

Step 1.1. Assembling the photometer (2 p)

To be circled and signed by a lab assistant after you raise the red card: (a) The photometer is correctly assembled. _____ (b) The photometer is not correctly assembled. The lab assistant provided the solution. _____	Marks
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Step 1.2: Preparation of solutions for measurement

1.2.1. Preparation of sodium hydroxide solutions

- 1.2.1. Fill in the Table 1.2.1 and detail your calculations only for the first line on the Answer Sheet below Table 1.2.1.! Indicate your final result for [NaCl] using the scientific notation with 3 significant figures (example: 1.21×10^{-5}). For calculated volumes of solutions A and B, indicate the result as a number with 2 significant figures. (4.5 p)

Table 1.2.1. Data for preparation of solutions I-III						Marks
SOLUTION/ EXPERIMENT LABEL	[NaOH] (M)	[NaOH] + [NaCl] (M)	[NaCl] (M)	V(A) (mL)	V(B) (mL)	
I	1.20×10^{-2}	3.00×10^{-2}				
II	9.60×10^{-3}	3.00×10^{-2}				
III	7.20×10^{-3}	3.00×10^{-2}				

Detailed calculation for the first line (1.5 p):

	Marks
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1.2.2. Preparation of CV solution

- 1.2.2. Show details of your calculations. Indicate your final result for $V_{ST}(CV)$ as a number without decimal places in the μL units. (1 p)

Detailed calculation:

	Marks
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Step 1.3. Measurement of the reaction rate

1.3.1. Voltage measurement for the blank sample (distilled water) (1.6 p)

<u>Checkpoint! To be filled and signed by a lab assistant after you raise the red card:</u> Voltage for the blank sample (distilled water) = _____ mV LED diodes changed: Yes No _____	
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Table 1.3.1. Measured values of U_{BL}		Marks
Measurement number	U_{BL} (mV)	
1		
2		
3		
Express the measurements in mV units without decimals.		

- 1.3.1. Calculate the average value ($\overline{U_{BL}}$) for the blank sample. Show details of your calculations and indicate your final result for $\overline{U_{BL}}$ as a number without decimal places in mV units.

	Marks
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1.3.2. Temperature in the laboratory

Before continuing, raise the red card for the lab assistant to record the temperature in the lab. The rate constant is temperature-dependent and it is very important to record the temperature.

Temperature in the lab = _____ °C	NO POINTS
Time: _____	
Lab assistant's signature: _____	

Step 1.4. Data collecting and processing

1.4.1. Calculation of the absorbances and $\ln(A_t)$ (11.8 p)

- Fill in the **Tables 1.4.1.** with the measured voltages in mV (without decimal places). Absorbances and $\ln(A_t)$ must be indicated as numbers with 4 decimal places.

Marks

- Detail your calculations **only for the first line of Table 1.4.1.** for Experiment I only.



Tables 1.4.1. Measured and calculated values.

Experiment I				
Data point	t (min)	U (mV)	A_t	$\ln(A_t)$
1.	0:00			
2.				
3.				
4.				
5.				
6.				
7.				
8.				

Experiment II				
Data point	t (min)	U (mV)	A_t	$\ln(A_t)$
1.	0:00			
2.				
3.				
4.				
5.				
6.				
7.				
8.				

Experiment III				
Data point	t (min)	U (mV)	A_t	$\ln(A_t)$
1.	0:00			
2.				
3.				
4.				

5.				
6.				
7.				
8.				

- Detail your calculation for the first line of Experiment I. Absorbances and $\ln(A)$ must be indicated as numbers with 4 decimal places.

	Marks
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1.4.2. Calculation of the k' (11.1 p)

- 1.4.2.1. Draw a graph of $\ln(A_t)$ versus t on the millimeter paper. Data points must be shown as $\times$, points used for the calculation of the slope must be shown as $\otimes$. Label the paper with your team/country's sticker.
- 1.4.2.2. Determine the reaction rate constants k' for each of the three measurements from **Graph 1.4.2.1**. Detail your calculations **only for the Experiment with solution I** on the Answer Sheet. Express the values in units min^{-1} having 4 decimal places and fill in the Table 1.4.2.

	Marks
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Table 1.4.2. Determined values of k'		Marks
SOLUTION/ EXPERIMENT LABEL	k' (min ⁻¹)	
I		
II		
III		

1.4.3. Calculation of the rate constant k (8 p)

Before continuing, raise the red card for the lab assistant to record the temperature in the lab. The rate constant is temperature-dependent and it is very important to record the temperature.

Temperature in the lab = _____ °C Time: _____ Lab assistant's signature: _____	NO POINTS
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- 1.4.3.1. Assuming the OH^- concentration remains constant during the experiment, calculate its concentration in the cuvette for all three experiments. Results must be given using the scientific notation with 1 decimal place (example: 1.2×10^{-5}).

	Marks
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- 1.4.3.2. Draw the graph on the millimeter paper using the data from **Table 1.4.2.** and concentrations of $[OH^-]$ in all three experiments from part 1.4.3.1. Elements on the graph that must be shown: axes with titles and values, all $k' - [OH^-]$ values, regression lines, data points must be shown as $\times$, points used for the calculation of the slope must be shown as $\otimes$. Label the paper with your team/country's sticker.
- 1.4.3.3. From **Graph 1.4.3.1.** calculate the value of the rate constant k . Show the details of your calculations. Final value of k that you report must be in $M^{-1} \text{ min}^{-1}$ units and have 2 significant figures.

Detailed calculation of k :

	Marks
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Problem 2. Velocity and Drag

(TOTAL MARKS: 43.5)

2.1 (3.2 p), 2.2 (0.4 p), 2.4 (14.4 p) and 2.5 (1 p).

Paper width (cylinder height) $L = 3.5$ cm		
angle θ (deg)	time t (s)	time t (s)
20	0	0
13		
8		
5		
3		
Pendulum period:		

Paper width (cylinder height) $L = 5.0$ cm		
angle θ (deg)	time t (s)	time t (s)
20	0	0
13		
8		
5		
3		
Pendulum period:		

Paper width (cylinder height) $L = 7.0$ cm		
angle θ (deg)	time t (s)	time t (s)
20	0	0
13		
8		
5		
3		
Pendulum period:		

Paper width (cylinder height) $L = 14.0$ cm		
angle θ (deg)	time t (s)	time t (s)
20	0	0
13		
8		
5		
3		
Pendulum period:		

Paper width (cylinder height) $L = 21.0$ cm		
angle θ (deg)	time t (s)	time t (s)
20	0	0
13		
8		
5		
3		
Pendulum period:		

angle θ (deg)	logarithm of the angle
20	3.00
13	
8	
5	
3	

Table 2.2. Measured values of the times for different paper cylinder dimension, at fixed angles 3 deg, 5 deg, 8 deg and 13 deg. (20 deg corresponds to initial condition at $t = 0$.) Bottom right Table: logarithm of the angles.

2.3. Calculate the period of oscillation using equation (2.2). Determine the relative error between the measured and the calculated period values (relative to calculated value). Show your calculations. (3 p)

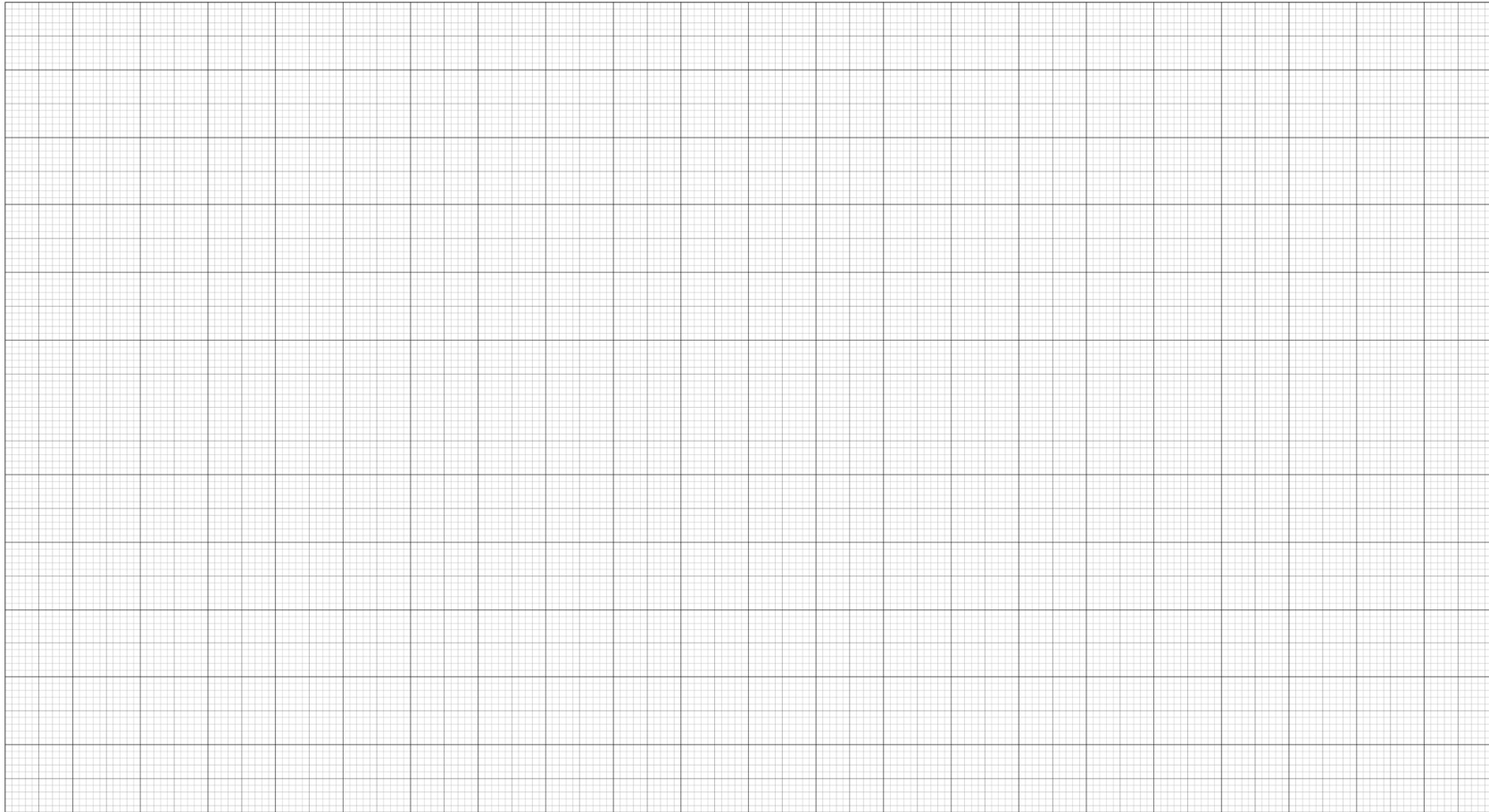
		Marks
Value of R	$R = \underline{\hspace{2cm}}$	
Calculated value of period T	$T = \underline{\hspace{2cm}}$	
Relative error between measured and calculated period T	relative error $T = \underline{\hspace{2cm}}$	
Calculations		

2.6. Given the measured values of the oscillation period T , is the drag caused by the paper cylinder negligible, or non-negligible? Show your calculations. Does the period T of the pendulum decrease or increase as the height of cylinder increases? (1 p)

		Marks
The drag is	☐ negligible	
	☐ non-negligible	
Period T	☐ decreases with height	
	☐ increases with height	
Calculations		

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2.7. Using the values from Table 2.2, plot time dependence of calculated values of the logarithm of the angle ($\ln \theta$), on graphing paper (Plot 2.1). Plot ALL data, without averaging. (6.5 p)



Plot 2.1. Graph of $\ln \theta$ -vs- t at fixed angles, for different paper cylinder heights L .

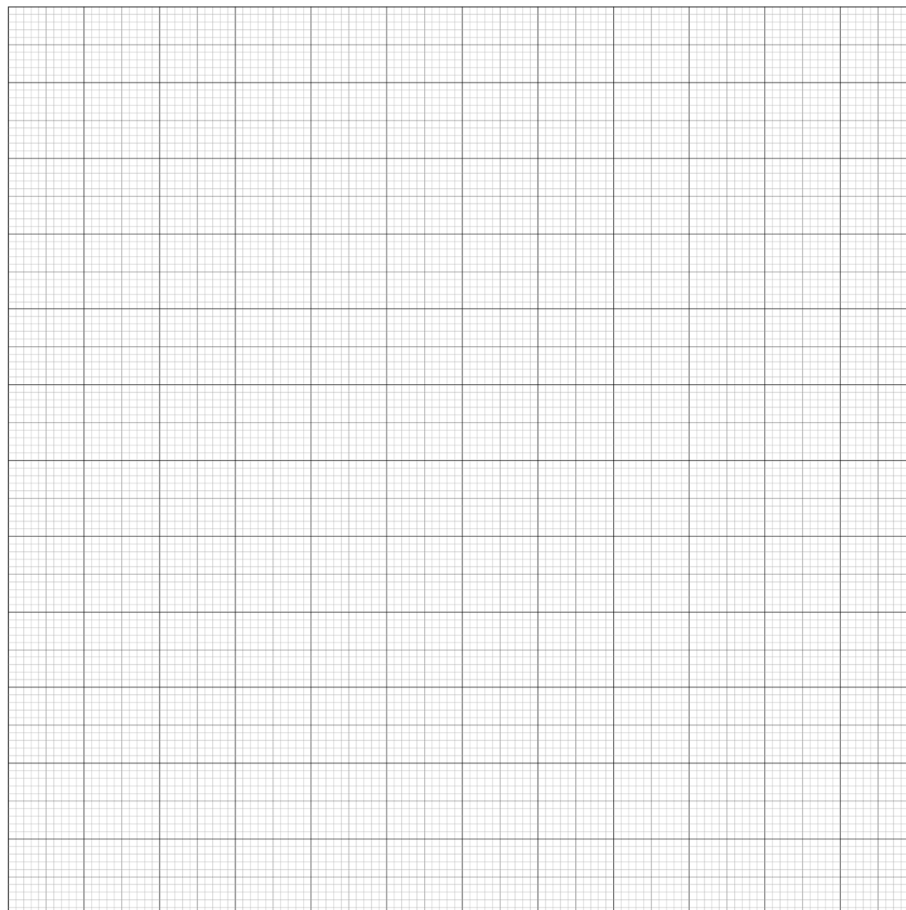
2.8. For each dimension L , draw a straight line that best follows the points, and from the slope of that line determine the damping factor γ to three significant figures. Fill in the first two columns of Table 2.3. with the obtained values. (7.5 p)

L (cm)	γ (1/s)	renormalized L_{ren} (cm)	Marks

Table 2.3.

2.9. For each value of the cylinder dimension L_i , determine the corresponding renormalized value L_{ren} , and enter the value in the last column of Table 2.3. (2.5 p)

2.10. On the graphing paper (Plot 2.2), plot values of γ as a function of the renormalized dimension L_{ren} . (2 p)



Plot 2.2. Dependence of γ on the renormalized dimension L_{ren} .

2.11. Using a ruler, draw the line of best fit on the graphing paper (Plot 2.2), and from its slope determine the value of constant C . Show your calculations. (2 p)

		Marks
Value of constant C	$C =$ _____	
Calculations		

Problem 3. Heat it up! Speed it up! Step it up!

(TOTAL MARKS: 40)

STEP 3.1. Complete the Table 3.1.1. by entering the respective treatment temperatures (t_1 for the lower temperature and t_2 for the higher temperature), volumes of expelled water for each temperature treatment for each tube and mean volumes of expelled water for each temperature treatment (V_1 for the mean at t_1 , and V_2 for the mean at t_2). If you have inconsistent volume, eliminate the tube with extreme minimum (outlier).

Table 3.1.1. Treatment temperatures and volumes of expelled water. (2 p)

	t_1	t_2
Temperature / °C		
Tube 1 Volume of expelled water / mL		
Tube 2 Volume of expelled water / mL		
Tube 3 Volume of expelled water / mL		
Tube 4 Volume of expelled water / mL		
Tube 5 Volume of expelled water / mL		
Mean volume of expelled water / mL Round to one decimal place		
Marks		

STEP 3.2.

3.2.1. Write down the Q_{10} coefficient that you obtained from the yeast fermentation experiment.
Round the number to one decimal place. (3 p)

The Q_{10} is:	Marks
When you calculate the Q_{10} , please call the lab assistant for a check-up. If it is correct, carry on with your assignment. If wrong, the lab assistant will give you the right answer, but you will lose 3 points.	Lab assistant signature

STEP 3.3.

Table 3.3.1. Write down the generation time of bacteria in YOUR rice bowl by using the Q_{10} coefficient obtained from the yeast experiment. In your case, you have a given generation time at 4 °C which is 8.0 h. **Round generation time to one decimal place and use this value for following calculations.** (1.5 p)

	Your results	Marks
Q_{10} coefficient		
Generation time (h) at 4°C	8.0	–
Generation time (h) at 14°C		
Generation time (h) at 24°C		
Generation time (h) at 34°C		

Table 3.3.2. Complete the missing data in in the Table below. Determine the number of divisions (n) that occur over a 24-hour period at each temperature. **Round the number of divisions to the whole number.** Use the obtained data to calculate the final numbers of bacterial cells (N) at each temperature. Write N in scientific format. (4 p)

Temperature (°C)	Generation time (h)	N_0	n – number of divisions in 24 h	N	Marks (n)	Marks (N)
4	8.0	500				
14		500				
24		500				
34		500				

STEP 3.4.

Answer the following questions.

3.4.1. (1 p)

Question	Answer	Marks
What was the mass of glucose in your labelled package?		

3.4.2. (1 p)

Question	Answer		Marks
Carefully look at the experimental setup from Step 3.1. and answer this: If the experiment had continued for another three hours, would the liquid have overflowed from the plastic syringe? Round the correct statement.	YES	NO	

3.4.3. Question: The experiment used expelled water as an indirect measure of yeast metabolism. Which of the following could have been another reliable way to measure the metabolic rate of yeast in the same experiment? **There can be one or more than one correct answers**, so mark it/them with an X. (1 p)

		Answer	Marks
A	Counting the number of yeast cells under a microscope before and after incubation.		
B	Measuring the amount of CO ₂ gas produced during fermentation.		
C	Measuring the amount of O ₂ gas produced during fermentation.		
D	Measuring the amount of lactic acid produced during fermentation.		
E	Measuring the amount of propionic acid produced during fermentation.		

3.4.4. In the tubes, the yeast underwent fermentation. However, this included an essential biochemical process, which is: There can be one or more than one correct answers, so mark it/them with an X. (1 p)

		Answer	Marks
A	Glycolysis, which requires oxygen.		
B	Glycolysis, which does not require oxygen.		
C	Citric acid cycle, which requires oxygen.		
D	Citric acid cycle, which does not require oxygen.		
E	Calvin cycle, which requires oxygen.		
F	Calvin cycle, which does not require oxygen.		

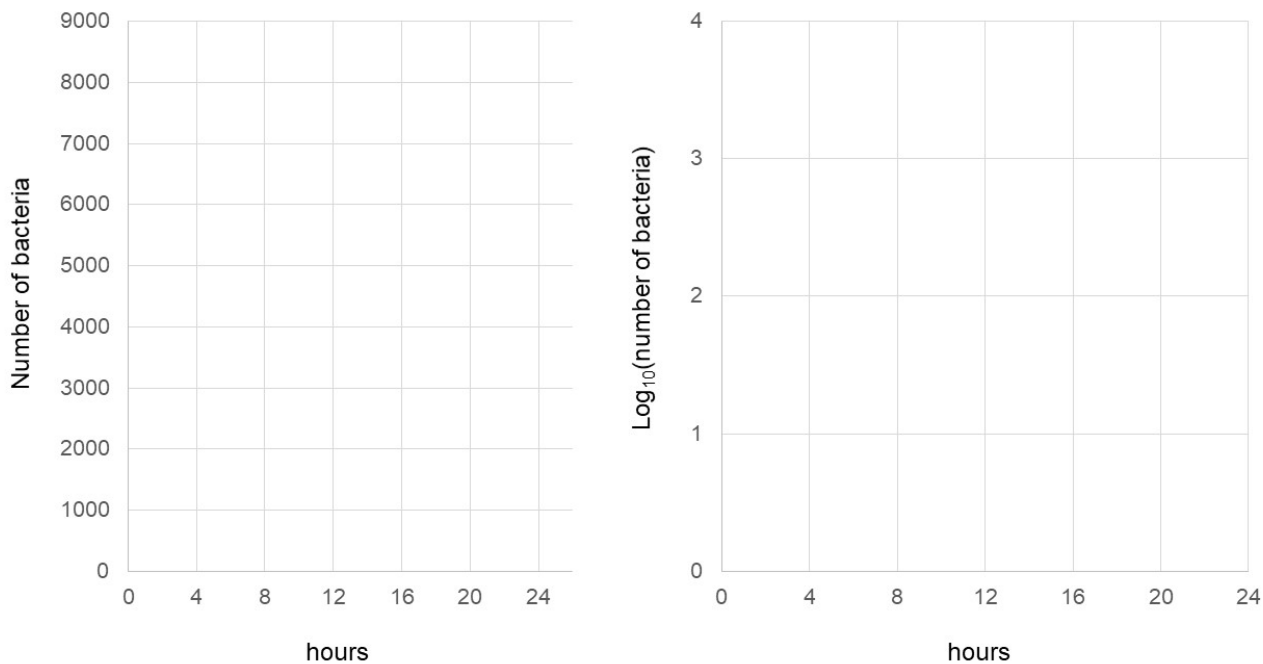
3.4.5. Question: A bakery wants to optimize dough fermentation for faster rising. Based on your knowledge gained in this experiment, and your previous knowledge, which of the following would be the best advice for the bakers? **There may be one or more than one correct answers**, so mark it/them with an X. (1 p)

		Answer	Marks
A	Increase the dough temperature to 30 °C.		
B	Increase the dough temperature to 60 °C.		
C	Add some artificial sweetener to the dough.		
D	Add some table sugar to the dough.		
E	Add a small amount of ethanol to pre-activate fermentation enzymes.		

3.4.6. Question: Examine the results obtained in **Table 3.3.2.** - the number of bacteria after 24 h at different temperatures. Assume that anything under a thousand bacteria in the bowl is safe to consume. Now answer the following questions (1 p):

		Answer		Marks
A	If the starting number of bacteria was 1000, would the food be OK to eat after one day if kept at 34 °C?	YES	NO	
B	If the starting number of bacteria was 10, would the food be OK to eat after one day if kept at 34°C?	YES	NO	
C	What if there was just a single bacterium in the bowl, would it be OK to eat after one day kept at 34°C?	YES	NO	
D	How about if there were no bacteria at all in the bowl, would it be OK to eat after one day kept at 34 °C?	YES	NO	

3.4.7. This will need some mathematical thinking, as well as artistic touch. Draw a curve of bacterial growth with following data: **generation time of 2 h**, and starting number of bacteria of 2 cells. (2 p)



	Marks
Graph 1	
Graph 2	

3.4.8. Question: Undeniable evidence that food quality is highly dependent on storage temperature and the initial number of bacteria is all around you—even if you haven't noticed it until now! :)

In the following table, **all statements are factually correct**, but your task is to mark with an "X" only those that conclusively prove that an **increase in storage temperature and the initial bacterial presence** in food are key factors affecting its safety. (1 p)

		Answer	Marks
A	Cheese kept in the refrigerator will be ok for some time, but it will spoil much faster if kept at warm place.		
B	Some types of milk can be kept in refrigerator, but will spoil pretty quickly if kept at warm place. On the other hand, some types of milk are kept in the supermarket outside the refrigerator, but still they do not spoil.		
C	If we keep canned food in the refrigerator, it will be good for a prolonged period of time.		
D	If we keep canned food outside the refrigerator, even in the summer, it will be good for a prolonged period of time.		
E	Obviously closed bag of chips can stay outside the refrigerator for months, and will be good to eat. But, if we open the bag of chips, eat from it and then keep it at our desk for few days, the chips is still good to eat.		
F	Similarly, if you open a jar of honey, take a spoonful and eat it, then use the same spoon to take another, and finally a third taste using the same spoon, you would have transferred many bacteria from your mouth into the honey. However, even after two weeks, the honey will not spoil. Even during the summer and outside the refrigerator.		

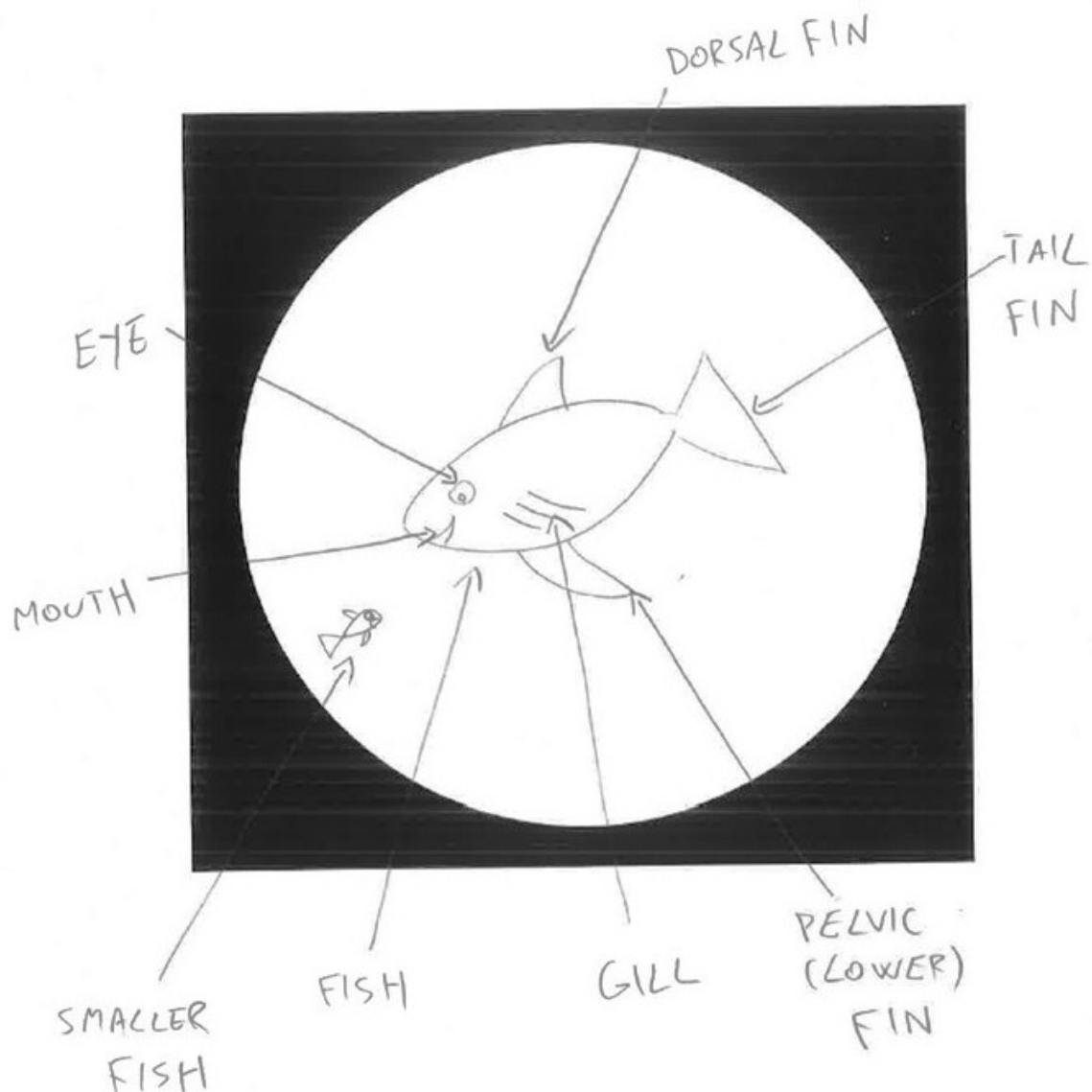
STEP 3.5.

3.5.1. Using the immersion objective, carefully examine your stained slide and draw what you observe. Draw a representative image within the circle (the field of view under the microscope). Be sure to include the important features, focusing on a field of view where both epithelial cells and bacterial cells are visible.

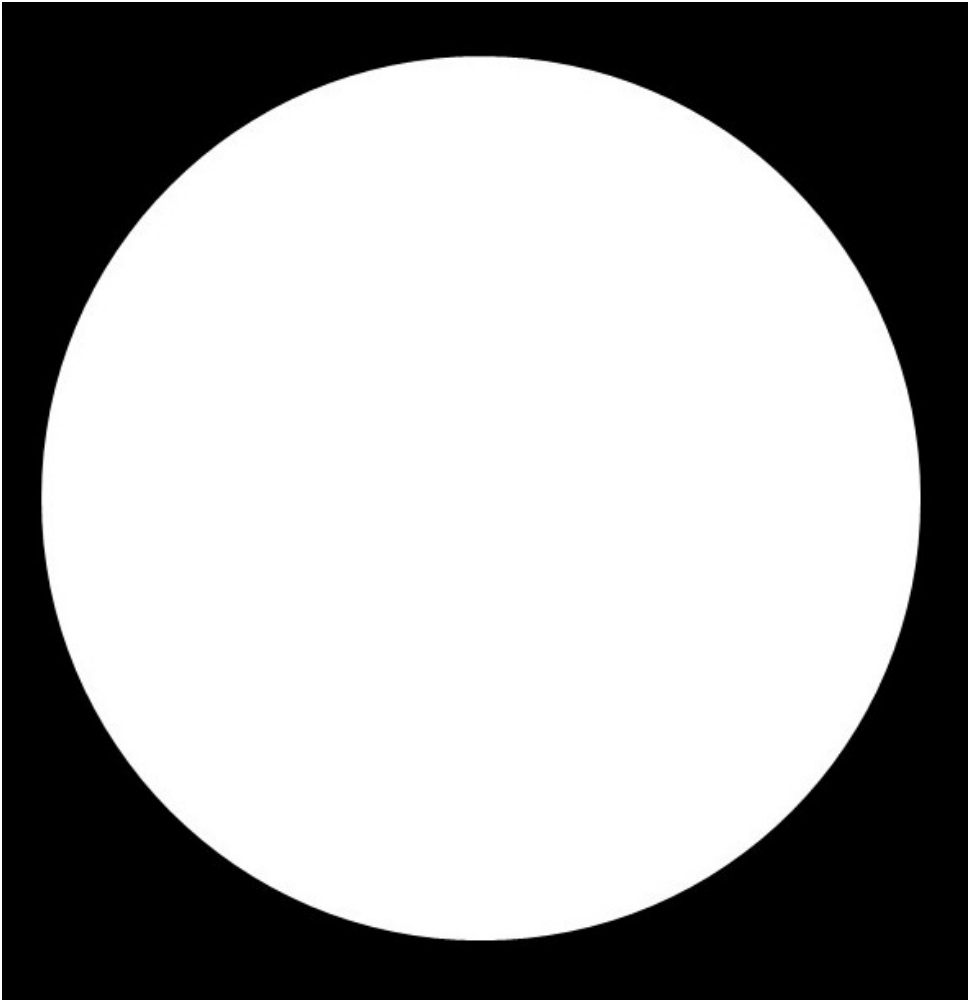
Label the key features on your drawing. Use arrows in your drawing to point to the important structures you should observe. (5.5 p)

Example:

Below is an example of how your drawing might look.



3.5.1. Representative image of your swab sample

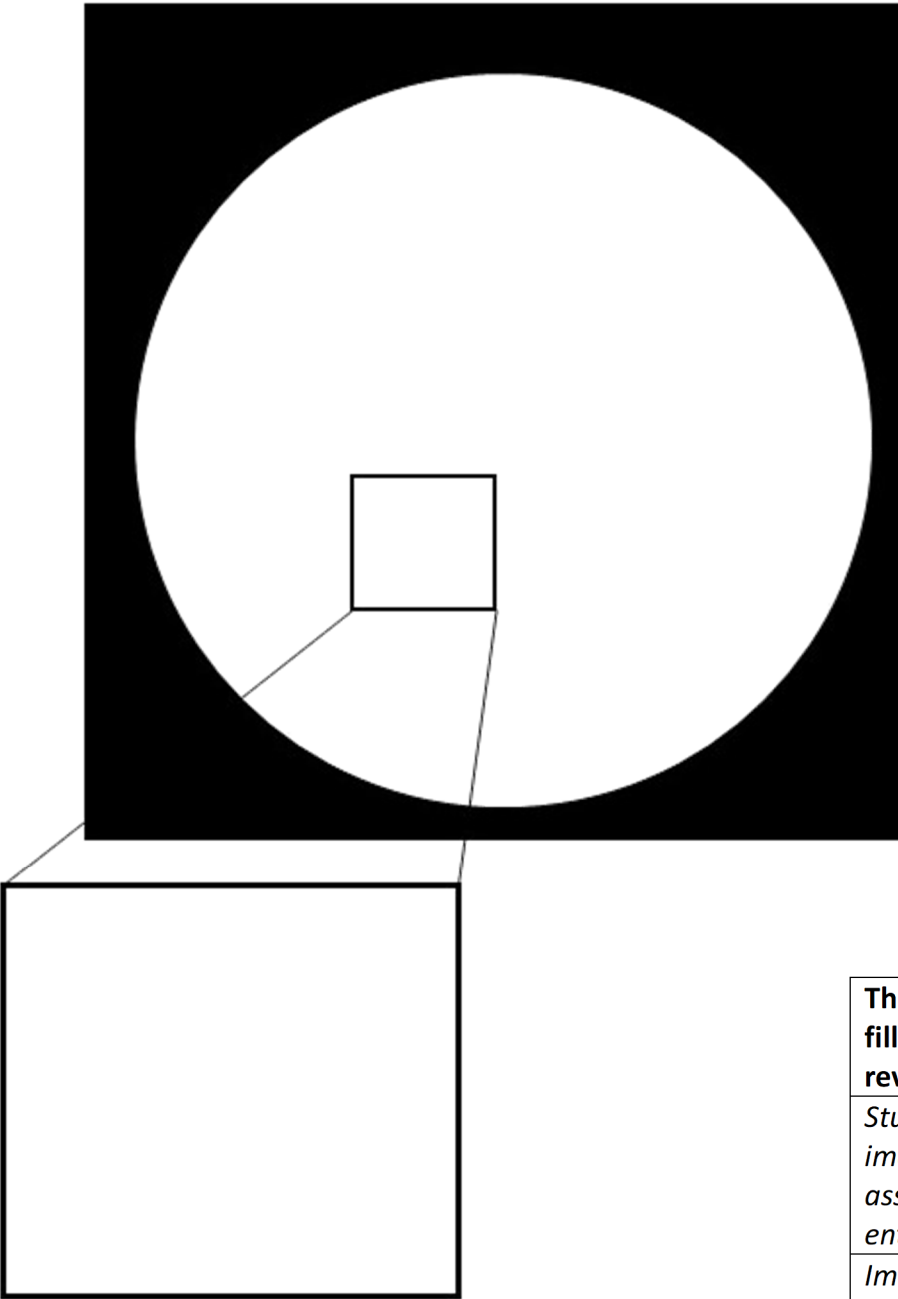


This Table is to be filled by the reviewers	Marks
Student found the image (if not, Lab assistant must enter 0 here)	
Image is drawn correctly	
Key features	

STEP 3.6. (3 p + 4 p + 5 p)

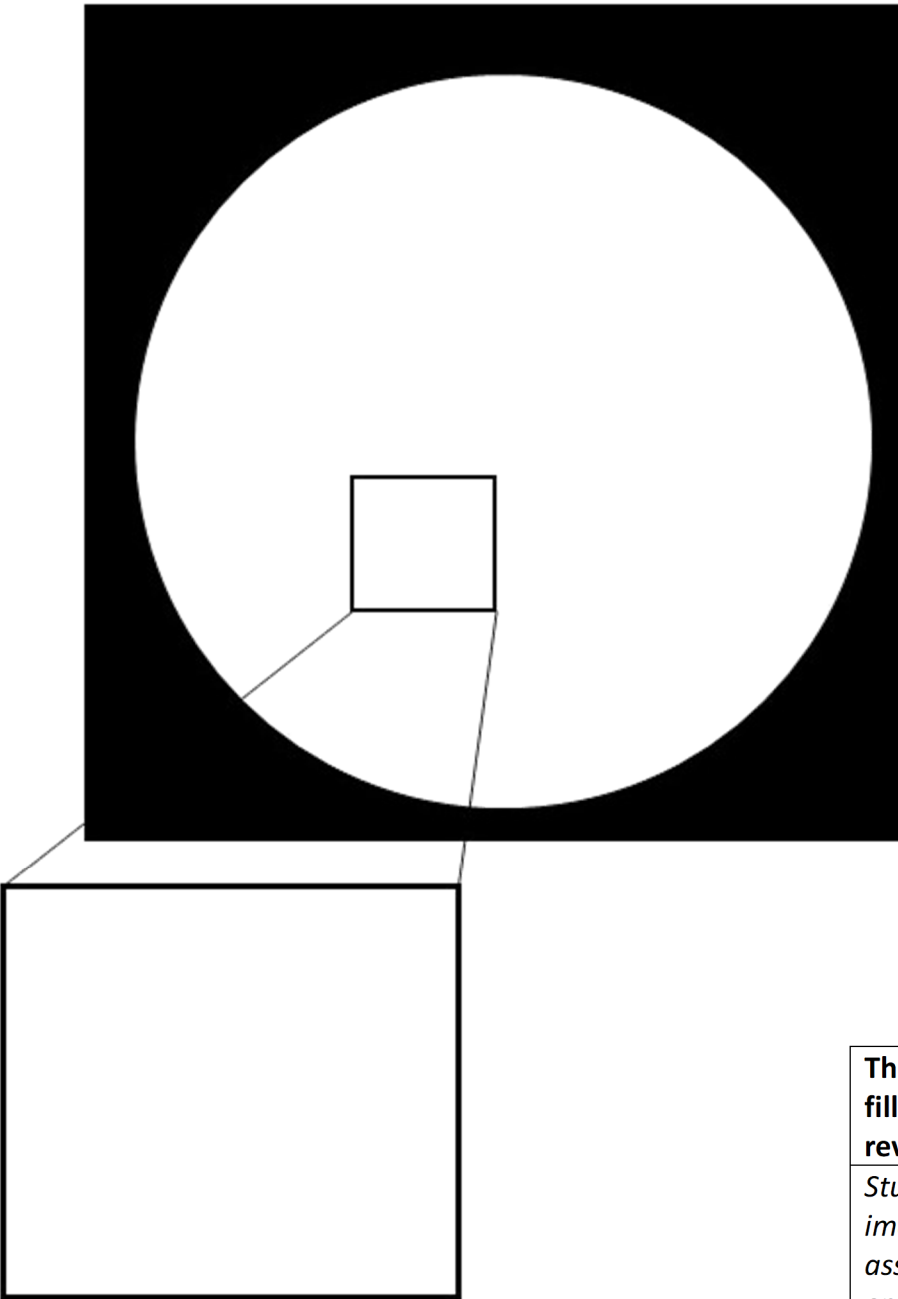
3.6.1. Draw representative image of slide **marked X1**. Mark key features, if there are any.

HINT – Some bacteria divide in a way that their daughter cells remain attached, forming distinctive clusters with a constant geometric pattern, where the number of bacteria is always the same. Can you identify such clusters in your sample? If so, draw an enlarged schematic image in the square beneath the field of view.



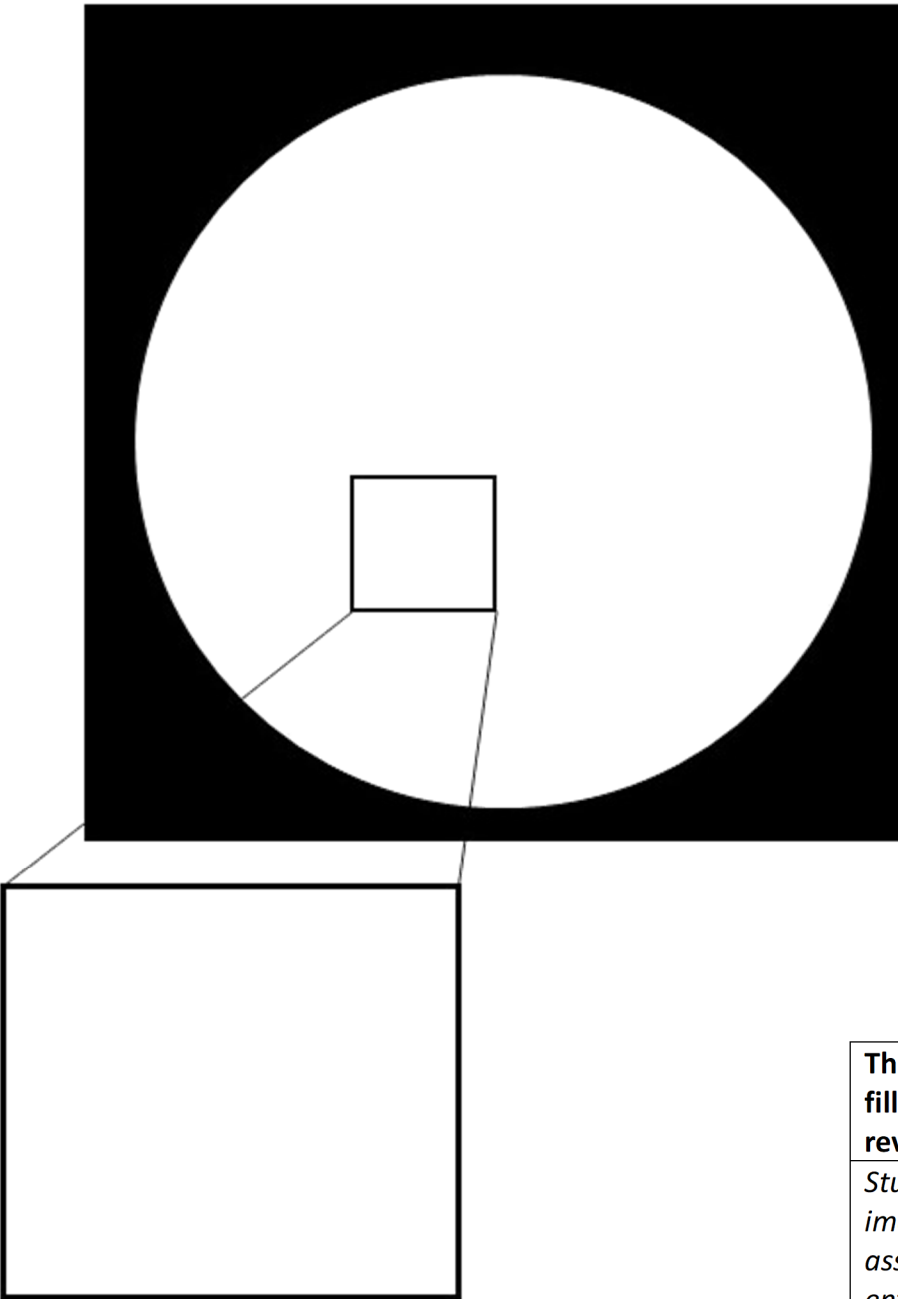
This Table is to be filled by the reviewers	Marks
<i>Student found the image (if not, Lab assistant must enter 0 here)</i>	
<i>Image is drawn correctly</i>	
<i>Key features</i>	

3.6.2. Draw representative image of slide **marked X2**. Mark key features, if there are any.
HINT – All the cells on this slide are of a single bacterial species, *Bacillus cereus*. But, although its the same species, there are two morphologically distinctive types of cells that you can see, if you look carefully. Can you find them and draw an enlarged schematic image in the square beneath the field of view?



This Table is to be filled by the reviewers	Marks
<i>Student found the image (if not, Lab assistant must enter 0 here)</i>	
<i>Image is drawn correctly</i>	
<i>Key features</i>	

3.6.3. Draw representative image of slide **marked X3**. Mark key features, if there are any.
HINT – We prepared a mix of different microorganisms on this slide. If you can identify morphologically different cells, please do draw an enlarged schematic image of them in the square beneath the field of view? While drawing, try to keep the cell size ratio the same as you see it in the field of view.



This Table is to be filled by the reviewers	Marks
<i>Student found the image (if not, Lab assistant must enter 0 here)</i>	
<i>Image is drawn correctly</i>	
<i>Key features</i>	

STEP 3.7.

3.7.1. Question: Take a good look at the image drawn at 3.5.1. If you know that nucleus is around 10 μm in diameter, could you estimate the average size of your bacterial cells? Mark the correct answer(s) with an X. (1 p)

Answer		Marks
Around 10 nm		
Around 100 nm		
Around 1 μm		
Around 0.01 mm		
Around 0.001 mm		

3.7.2. Question: Observing the slide **X2** you should have been able to see two types of morphologically different bacterial cells. What do you think the difference was? Mark the correct answer(s) with an X. (1 p)

Answer		Marks
They are from two different bacterial species.		
Some cells have air bubbles inside them.		
Some cells have a feeding vacuole inside them.		
Some cells have enlarged nucleus due to division in progress.		
Some cells have endospores inside them.		

3.7.3. Question: Observing the slide **X3** you should have been able to see several types of morphologically different cells. What do you think what species they were? Mark the correct answer(s) with an X. (1 p)

Answer		Marks
Bacterium <i>Staphylococcus aureus</i> , bacterium <i>Olympiadus zagrabiensis</i> and yeast <i>Saccharomyces cerevisiae</i> .		
Bacterium <i>Staphylococcus aureus</i> , bacterium <i>Bacillus megaterium</i> and protozoan <i>Paramecium caudatum</i> .		
Bacterium <i>Staphylococcus aureus</i> , bacterium <i>Staphylococcus argentum</i> and bacterium <i>Staphylococcus cuprum</i> .		
Bacterium <i>Staphylococcus aureus</i> , bacterium <i>Bacillus megaterium</i> and yeast <i>Saccharomyces cerevisiae</i> .		
Bacterium <i>Bacillus cocoideus</i> , bacterium <i>Bacillus megaterium</i> and protozoan <i>Euglena viridis</i> .		
Yeast <i>Saccharomyces cerevisiae</i> , yeast <i>Saccharomyces cocoidensis</i> and yeast <i>Saccharomyces bacillae</i> .		